

Introduction to Professional Practices in IT

CS4001 – Professional Practices in IT

Asif Fazwani

Software Contracts & Licenses

Contracts

- A **contract** is a legally enforceable **agreement between two or more parties**.
- Parties can be **natural persons** (individuals) or **legal persons** (companies, organizations).

A freelancer developing a website via **Upwork** enters a contract where they provide development work in exchange for payment both parties give consideration.

Essentials of a Valid Contract

- Offer and Acceptance
- **Competency of Parties** (age & mental capacity)
- **Consideration** (something of value exchanged)
- **Intention to create legal relations**

LICENSE AGREEMENTS

- A license agreement is a legal contract between a software vendor and a customer.
- It defines how software can be used, distributed, and maintained.
- When buying software, customers purchase usage rights, not ownership of the software itself.

Common Types of License Agreements

1. Single-User License

- Allows use of **one copy on one computer**.
- Common for **consumer software** (e.g., games, photo editing tools).

2. Concurrent User Licence

- Software installed on a **server**, accessible by multiple users simultaneously (up to a limit).
- Common for **corporate environments**.

Common Types of License Agreements

3. Site License

- Allows **unlimited installations** within a specific location or organization.

Example: Universities often use *MATLAB Campus-Wide Licence* or *Microsoft Azure Dev Tools for Teaching*

Vendor Concerns in Licensing

- Protect intellectual property rights – ensure ownership remains with vendor.
- Control software usage – require extra fees for broader use.
- Ensure recurring revenue – through maintenance, updates, or subscriptions.
- Limit liability – avoid responsibility for defects or data loss.

Duration of Licenses

- Perpetual license: Valid forever (common in retail software).
- Term-based license: Fixed duration (annual or monthly).
- Marketing/distribution license: Usually for a fixed contract period.

Copyright Assignment

- The copyright owner can transfer full ownership through written assignment.
- The new owner gains all rights previously held.
- Example: When Microsoft acquired Activision Blizzard (2023), it gained ownership of all associated game IPs (e.g., Call of Duty).

Outsourcing

- Outsourcing (or facilities management) is when an organization delegates planning, management, and operation of certain functions to an external supplier.
- Common in both private and public sectors.
- Goal: improve efficiency and focus on core competencies.

Tesla outsources chip production to **TSMC** to focus on electric vehicle design and AI systems.

Netflix outsources cloud infrastructure to **Amazon Web Services (AWS)** to focus on streaming and content creation.

Key Components of IT Outsourcing Contracts

Outsourcing agreements are **complex and context-dependent**. Important contract elements include:

1. Performance monitoring and management.
2. Handling unsatisfactory performance
3. Transfer of assets
4. Staff transfers
5. Audit & IPR rights

Benefits of Outsourcing

- Access to specialized skills and advanced technology.
- Cost reduction and operational efficiency.
- Flexibility and scalability.
- Strategic focus on core business areas.
- Faster innovation through expert collaboration.

Fixed Price Contracts for Bespoke Systems

- **Definition:** A bespoke (tailor-made) system is a software or IT system custom-built to meet a specific organization's requirements.
- **Contrast:** Unlike off-the-shelf software, bespoke systems are unique and designed for a particular user or organization.
- **Example:**
 - Small: Customized spreadsheet with automation macros.
 - Large: Global enterprise systems with thousands of PCs and specialized databases.

Why Fixed Price Contracts?

- Fixed price contracts are used when:
 - **Scope is well-defined** at the start.
 - The client wants **cost certainty**.
 - The supplier assumes responsibility for **delivering within budget and deadline**.
- Goal: Deliver a complete working system **at an agreed price**, regardless of actual cost to supplier.

Typical Structure of a Fixed Price Contract

A contract for a bespoke system usually includes **three main parts**:

- **Short Agreement**
 - Signed by both parties.
 - Identifies the parties involved.
 - States that **previous discussions or documents are not part of the contract**.
 - References terms and annexes.
- **Standard Terms and Conditions**
 - General legal and commercial terms used by the supplier.
 - Cover issues like warranty, liability, payment, and ownership rights.
- **Schedules or Annexes**
 - Specify **system requirements, delivery milestones, payment schedule, and support arrangements**.

Simpler Contracts for Small Projects

- Not all bespoke systems need detailed multi-part contracts.
- **Small-scale projects** can often be handled through:
 - An **exchange of letters** or
 - A **simple memorandum of understanding (MoU)**.

Example:

- A startup hiring a freelance developer to build a custom CRM dashboard for PKR 100,000 might use a one-page letter agreement.

Why These Contracts Matter

- Even though large bespoke system contracts are **less common**, they are **important for illustrating core IT contract issues**, such as:
 - Scope definition
 - Risk allocation
 - Change management
 - Liability and warranty
 - Delivery acceptance

Benefits of Fixed Price Contracts

- **Budget certainty** – the client knows the total cost upfront.
- **Encourages efficiency** – supplier is motivated to deliver quickly and within budget.
- **Reduced administrative complexity** – single cost negotiation.
- **Predictable planning** – easier to manage financing and timelines.

Risks and Challenges

- **Supplier risk:** Any scope change or underestimated cost reduces profit.
- **Client risk:** If requirements are unclear, the supplier may “cut corners” to stay within budget.
- **Limited flexibility:** Difficult to adapt to new needs mid-project.
- **Disputes over deliverables:** What qualifies as “completion” can be contested

What is to be Produced

- It is clearly necessary that the contract states what is to be produced.
- Requirements specification
- It is important that the reference to the requirements specification identifies that document uniquely; normally this will mean quoting a date and issue number.
- **Problem:** Any changes needed during the contract life

What is to be delivered

The following is a non-exhaustive list of possibilities:

- Source Code
- Command files/ batch files for building the executable code from the source and for installing it.
- Documentation of the design of the code.
- Reference manuals, training manuals and operations manuals.
- Software tools to help maintain the code.
- User training
- Training for the client's maintenance staff;
- Test Cases and test results.
- Deployment.

Ownership of rights

- It is important for the contract to state precisely who is to own these rights and what legal rights are being passed by the software house to the client under the contract.
- Ownership in physical items will usually pass from the software house to the client, but other intangible rights (known as intellectual property rights) making more problems.
- Software is potentially protectable by a number of intellectual property rights, such as copyright, design rights, confidentiality and trademarks.

Confidentiality and data protection

- When a major must software system is being developed, Client have to pass confidential information about its business operations to the software house.
- On the other hand, the software house may not want the client to disclose to other details of the program content.
- Non of the parties would like the other to disclose its secrets.
- It is usual in these circumstances for each party to promise to maintain the confidentiality of the other's secrets, and for express terms to that effect to be included in the contract.

Payment terms

A pattern of payments such as the following must be included:

- An initial payment of 15% of the contract value becomes due on signature of the contract. (Requirement gathering)
- Further stage payments become due at various points during the development, bringing the total up to 65%. (Development and testing)
- A further 25% becomes due on acceptance of the software. (Deployment)
- Final 10% becomes due at the end of the warranty period. (Beta Passed)

Calculating payments for delays and changes

- The contract must specify the process by which these extra payments are to be calculated.
- The contract should make provision for payments to compensate for the wasted effort, incurred, for example, when the client fails to provide information on a due date or when changes are requested that result in extra work.
- Delay payments and payments for variations to the original requirements are the most common cause of contractual disputes.

Penalty clauses

Delays in delivering working software are common; it might therefore be expected that contracts for the supply of software would normally include such a penalty clause. There are three reasons for this:

- Suppliers are very reluctant to accept penalty clauses and anything stronger than the example quoted above is likely to lead to reputable suppliers refusing to bid.

Example:

Suppose a government agency contracts a software company to deliver a tax management system within 12 months. The client insists on a penalty of **\$10,000 per week of delay**.

The supplier may **refuse to bid**, reasoning that unpredictable delays (such as scope changes or client-side testing delays) could easily add weeks or months. Reputable companies might decide it's too risky and decline to participate, leaving only low-quality bidders who are desperate for contracts.

Penalty clauses

- If the contract is to include penalty clauses, the bid price is likely to be increased by at least half the maximum value of the penalty.

Example:

Imagine the contract includes a **maximum penalty of \$200,000** for late delivery. To cover the potential risk, the supplier might increase their bid by **\$100,000 (half of the = maximum penalty)**. So instead of bidding \$500,000 for the software project, the supplier bids **\$600,000**, just in case delays occur and penalties apply. Thus, the client pays more upfront, even if no delay happens.

Penalty clauses

- If the software is seriously late and penalties approach their maximum, there is little incentive for the supplier to complete the work since they will already have received in stage payments as much as they are going to get.
- **Example:**

Suppose a supplier is contracted to deliver software worth **\$1 million**, paid in **stage payments** (e.g., \$800,000 received so far) and with a **maximum penalty of \$200,000** for late delivery.

If the project is very late and penalties hit the \$200,000 cap, the supplier effectively gets **no further payment** for finishing the work, they've already received all the money they're going to get.

At this point, the supplier might decide **not to prioritize** completion, focusing instead on other profitable projects, leaving the client with an incomplete system.

Obligations of the client

Client usually has some responsibilities to help get the work done smoothly.

These might include:

- Sharing relevant documents about their business or the system's environment.
- Giving access to the right staff members.
- Providing computers and equipment for development and testing.
- Offering workspace, phone, and administrative support if work is done at their location.
- Setting up data communication as needed.
- These obligations and deadlines are usually listed in the contract. If the client doesn't meet these, they might have to pay for any delays.

Standards and methods of working

- The supplier is likely to have company standards, methods of working, quality assurance procedures, and so on, and will normally prefer to use these.
- More sophisticated clients will have their own procedures and may require that these be adhered to. In some cases, the supplier may be required to allow the client to apply quality control procedures to the project. The contract must specify which is to apply.

Project meeting

- Regular progress meetings are essential to the successful completion of a fixed price contract, and it is advisable that standard terms and conditions require them to be held.
- The minutes of progress meetings, duly approved and signed, should have contractual significance in that they constitute evidence that milestones have been reached (so that stage payments become due) and that delay payments have been agreed.

Project managers

- Each party needs to know who, of the other party's staff, has day-to-day responsibility for the work and what the limits of that person's authority are.
- The standard terms and conditions should therefore require each party to nominate, in writing, a project manager.
- The project managers must have at least the authority necessary to fulfil the obligations that the contract places on them.
- It is particularly important that the limits of their financial authority are explicitly stated, that is, the extent to which they can authorize changes to the cost of the contract.

Acceptance Procedures

- Acceptance procedures are the criteria by which successful completion of the contract is judged.
- The essence of the acceptance procedure is that the client should provide a fixed set of acceptance tests and expected results and that successful performance of these tests shall constitute acceptance of the system.
- The tests must be provided at or before the start of the acceptance procedure; within reason, there may be as many tests as the client wishes, but extra tests cannot be added once the test set has been handed over. (Requirement gathering)
- The purpose of this restriction is to ensure that the acceptance procedure can be completed in reasonable time.

Warranty and maintenance

- Once the product has been accepted (Deployment done), beta testing required, typically 90 days. Any errors found in the software and reported within this period will be corrected free of charge. This clause is usually part of the contract.
- Once the warranty period is over, the supplier may offer, or the client demand, that maintenance will continue to be available on request (Support). Since such maintenance is likely to involve enhancement of the software rather than simply correction of faults, the resources required are unpredictable, the client almost certainly does not know what enhancements will be required in two years' time.
- For this reason, a fixed price for the maintenance will not be appropriate. Maintenance will therefore usually be charged on a time and materials basis

Inflation

- In long-term projects or project commitments for maintenance, the supplier will wish to ensure protection against the effects of unpredictable inflation. For that inflation clause added which allows charges to be increased in accordance with the rise in costs.
- The clause should state how often (once a year, twice a year) charges can be increased and how the effect on the overall price is to be calculated.

Termination of contract

There are many reasons to terminate a contract before it has been completed.

- It is essential to make provision in contract for terminating the work in an agreeable manner. This usually means that the supplier is to be paid for all the work carried out up to the point where the contract is terminated, together with some compensation for the time needed to redeploy staff on other revenue earning work.

Software Arbitration

- Contracts often contain a clause that in the event of a dispute that they cannot solve themselves, the parties agree to accept the decision of an independent arbitrator.
- In the case of computer-related contracts, arbitrator is to be appointed either by the President of the BCS or by the President of the IEE. Both bodies maintain lists of qualified arbitrators who have the necessary technical understanding.

CONSULTANCY AND CONTRACT HIRE

CONTRACT HIRE:

- Contract hire is an arrangement in which the supplier agrees to supply the customer with the services of a certain number of staff at agreed daily or hourly charge rates.
- The customer takes responsibility for managing the staff concerned. Either party can terminate the arrangement at short notice.

CONTRACT HIRE

- The supplier's responsibility is limited to providing suitably competent people and replacing them if they become unavailable or are unsuitable by the client.
- The contract for such an arrangement is usually simple. Payment is based on a fixed rate for each day worked. the rate depends on the experience and qualifications of the staff.
- Ownership of intellectual property rights generated in the course of the work may be needed to be addressed.

CONSULTANCY

- Consultants are experts who are called in by an organization to assess some aspect of its operations or its strategy and to make proposals for improvements.

Consultancy Project:

- A consultancy project involves providing expert advice or reports for a fixed price.
- These projects typically have simpler contracts compared to fixed-price contracts due to lower financial stakes and the challenge of objectively assessing the quality of reports.

Approaches:

- Simplified Contracts: Used because financial risks are minimal for both parties.
- Reliance on Reputation: Ensures quality through the supplier's desire to maintain their professional reputation.

Key Aspects of a Consultancy Contract

Confidentiality:

- Consultants gain access to sensitive company information.
- Safeguards are necessary to prevent misuse for personal profit.

Terms of Reference:

- Contracts must clearly define the scope of the consultancy work.
- Disagreements may arise if consultants want to expand beyond the original scope.

Key Aspects of a Consultancy Contract

Liability:

- Consultants often limit liability for losses caused by their advice.
- Clients may demand professional liability insurance to mitigate risks.

Control Over Final Report:

- Contracts usually require a draft report to be reviewed by the client.
- Clients can request changes, after which the consultant submits the final version.

Time and Materials Contracts

Definition:

A **Time and Materials (T&M)** contract is a hybrid between a **contract hire** and a **fixed-price** agreement.

Key idea:

Supplier is paid for **actual work done** — labour hours + materials.

Costs are open-ended but often have an **upper limit (“cap”)**.

Also known as: *Cost-plus* or *resource-based* contracts.

Analogy:

Like hiring a repair team by the hour — you pay for the time and resources used, not a fixed total.

When T&M Contracts Are Used

- Work scope not yet clearly defined.
- Project involves exploration, prototyping, or R&D.
- Requirements are expected to evolve.
- Client wants flexibility rather than rigid deliverables.
- Supplier and client agree on hourly/daily rates and expenses.

Advantages and Disadvantages

- **Advantages (for client):**
Flexibility in scope and direction.
Transparent cost tracking.
Better collaboration during development.
- **Disadvantages (for client):**
No total cost certainty.
Requires strong monitoring to avoid overruns.
Risk of inefficiency if supplier productivity drops.